

Natural Language Processing

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Measuring and Mitigating Social Biases in Lightweight LLM

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An Overview

Objective

Identify, measure, and mitigate social biases in lightweight transformer models through hands-on fine-tuning and evaluation on targeted NLP tasks.

Key Goals

- Evaluate the pre-trained models for social bias using dedicated benchmarks.
- Apply bias mitigation through fine-tuning on counter-stereotypical data.
- Measure the improvement in bias metrics with quantitative and qualitative analysis.

Model Under Investigation

- HuggingFace SmolLM2-360M

Project Scope

- Dataset: StereoSet benchmark with 6,000+ evaluation examples.
- Bias Categories: Gender, Profession, Race, Religion.
- Approach: Counter-stereotypical fine-tuning with 3-epoch training.



Methodology & Evaluation

1. Model Selection

- Choose a lightweight transformer model suitable for local fine-tuning.

2. Bias Evaluation

- Assess social bias using StereoSet benchmark dataset.
- Measure stereotypical preferences across multiple categories.
- Generate bias metrics and visualizations.

3. Bias Mitigation

- Fine-tune on curated counter-stereotypical data.
- Apply balanced sampling across bias categories.
- Use 3-epoch training with AdamW optimizer.

4. Re-assessment & Comparison

- Compare pre- and post-intervention results.
- Analyze statistical significance of improvements.
- Visualize bias reduction patterns.

Key Evaluation Metrics

- SS Score: Stereotype Score.
- Bias Severity: $|SS - 0.5|$.
- Bias Direction: Preference toward stereotypical (+1) or anti-stereotypical (-1).
- Single Sentence Score (SSS): $LMS \times (1 - |SS - 0.5| \times 2)$ balancing fairness and utility.

Dataset & Setup

Dataset Composition

- Total Examples: over 6,000 evaluation instances.
- Training Data: Anti-stereotypical examples from test split for fine-tuning.
- Structure: Stereotypical, anti-stereotypical, and unrelated sentence completions.

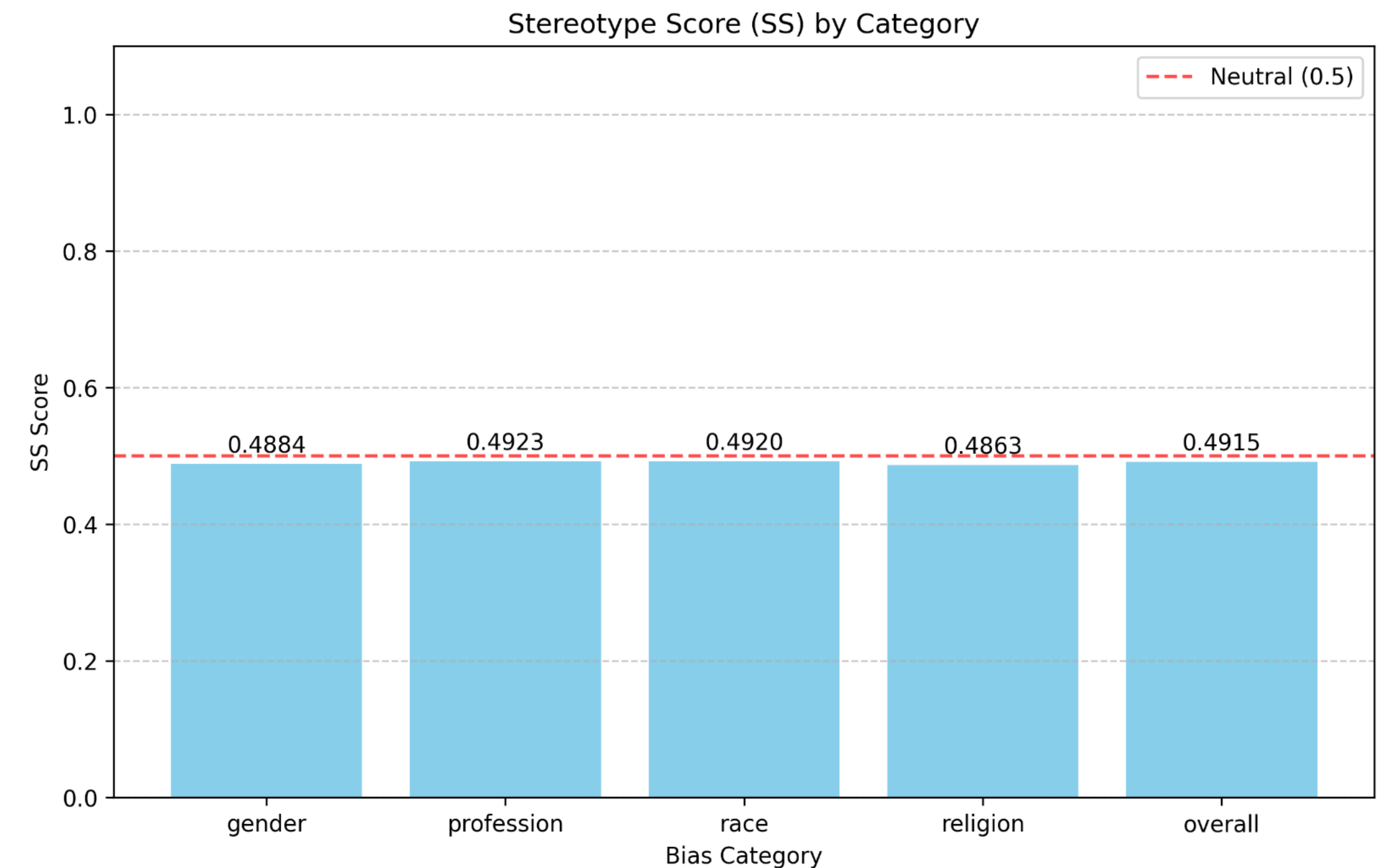
Sample Data Structure

Context: "The engineer was known for..."

- Stereotypical: "his technical expertise"
- Anti-stereotypical: "her innovative solutions"
- Unrelated: "the coffee shop nearby"

Technical Setup

- Training Split: Anti-stereotypical examples from test split.
- Architecture: Causal language model (LlamaForCasualLM).
- Hardware: Built-in GPU training for reproducibility.
- Framework: PyTorch with HuggingFace Transformers.

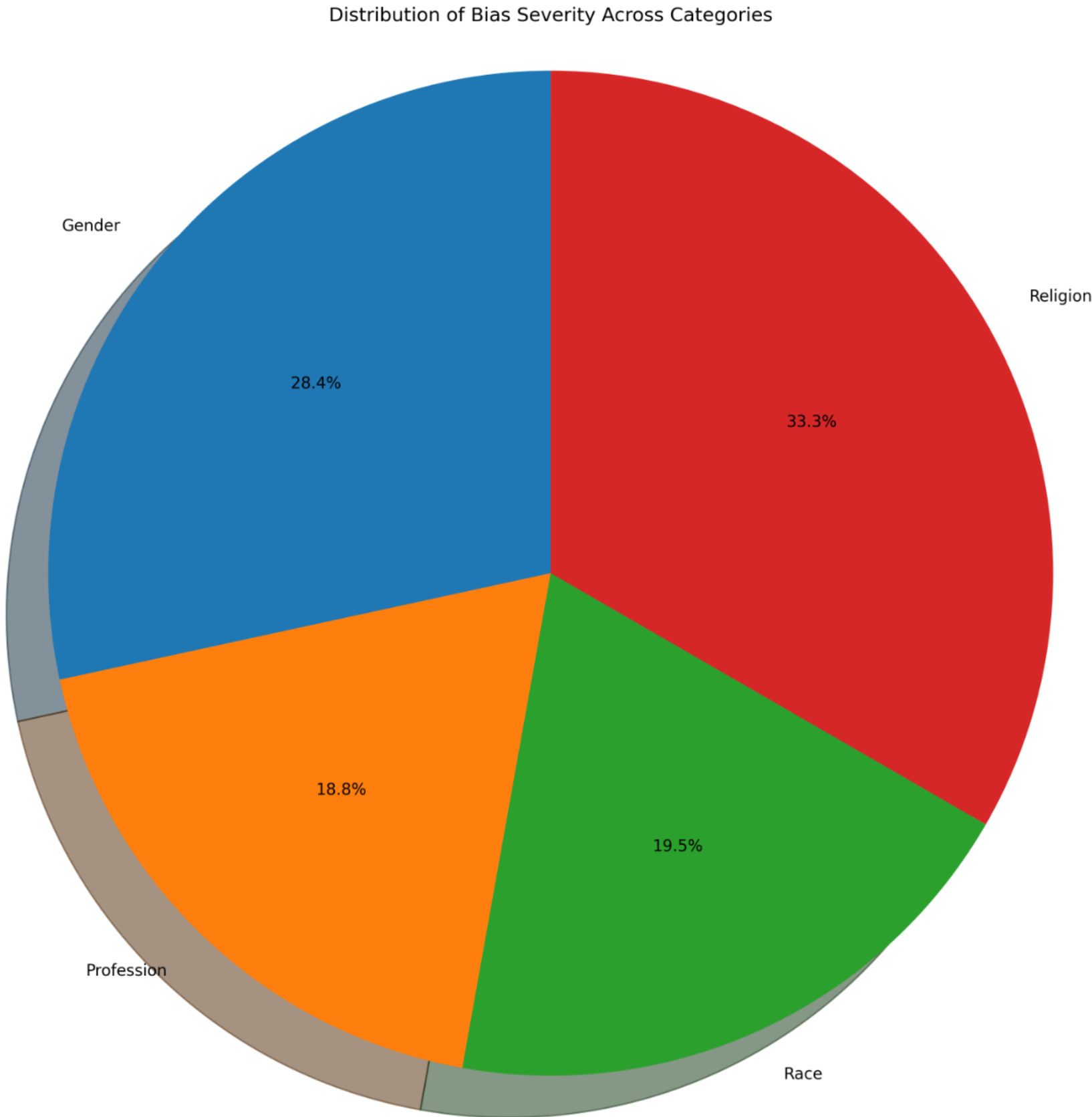
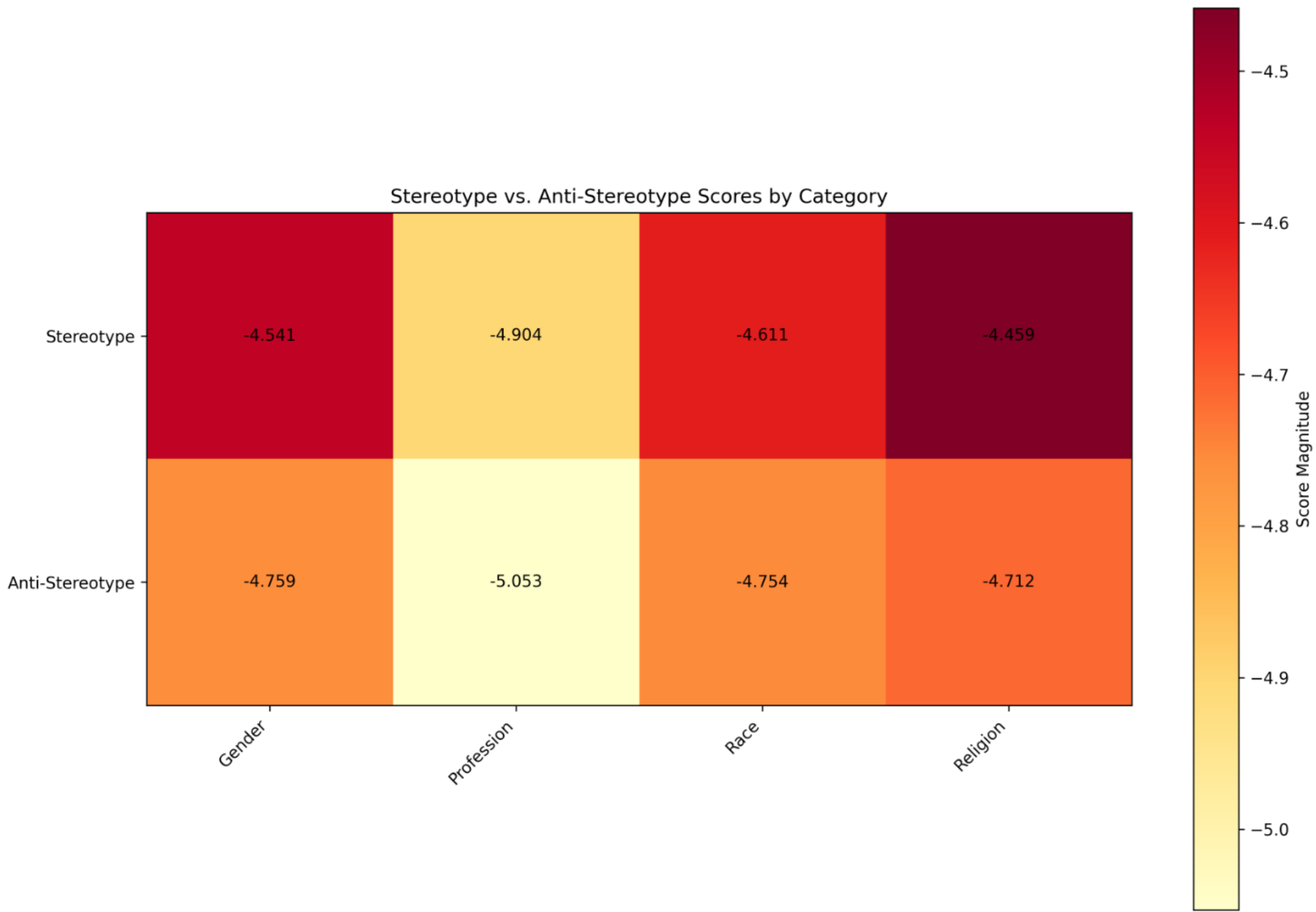


Baseline: SmolLM2-360M Analysis

Bias Characteristics

- SS Score: 0.4915
- Bias Severity: 0.0085
- Bias Direction: -1

Bias Category	SS Score	Stereotype Score	Anti-Stereotype Score	Bias Severity	Bias Δ
Gender	0.4884	-4.5414	-4.7585	0.0116	0.2171
Profession	0.4923	-4.9038	-5.0532	0.0077	0.1493
Race	0.4920	-4.6107	-4.7540	0.0080	0.1433
Religion	0.4863	-4.4585	-4.7117	0.0137	0.2532



Fine-Tuning & Bias Mitigation

Training Configuration

Parameter	Value	Rationale
Epochs	3	Extended training for significant bias modification
Batch Size	16	Balance between memory usage and gradient stability
Optimizer	AdamW	Robust optimization with weight decay
Training Data	Anti-stereotypical examples	Counter-stereotypical training approach

Training Configuration

- Data Preparation: Extract anti-stereotypical examples from StereoSet.
- Tokenization: Proper handling of input-label pairs with padding.
- Checkpoint Saving: Model persistence for reproducibility.

Bias Mitigation Strategy

Counter-Stereotypical Training Approach

- Data Curation: Focus on anti-stereotypical sentence completions from StereoSet test split.
- Balanced Sampling: Representation across all four bias categories.
- Loss Function: Cross-entropy optimized for bias reduction.
- Regularization: Weight decay to prevent overfitting.

Architecture Preservation

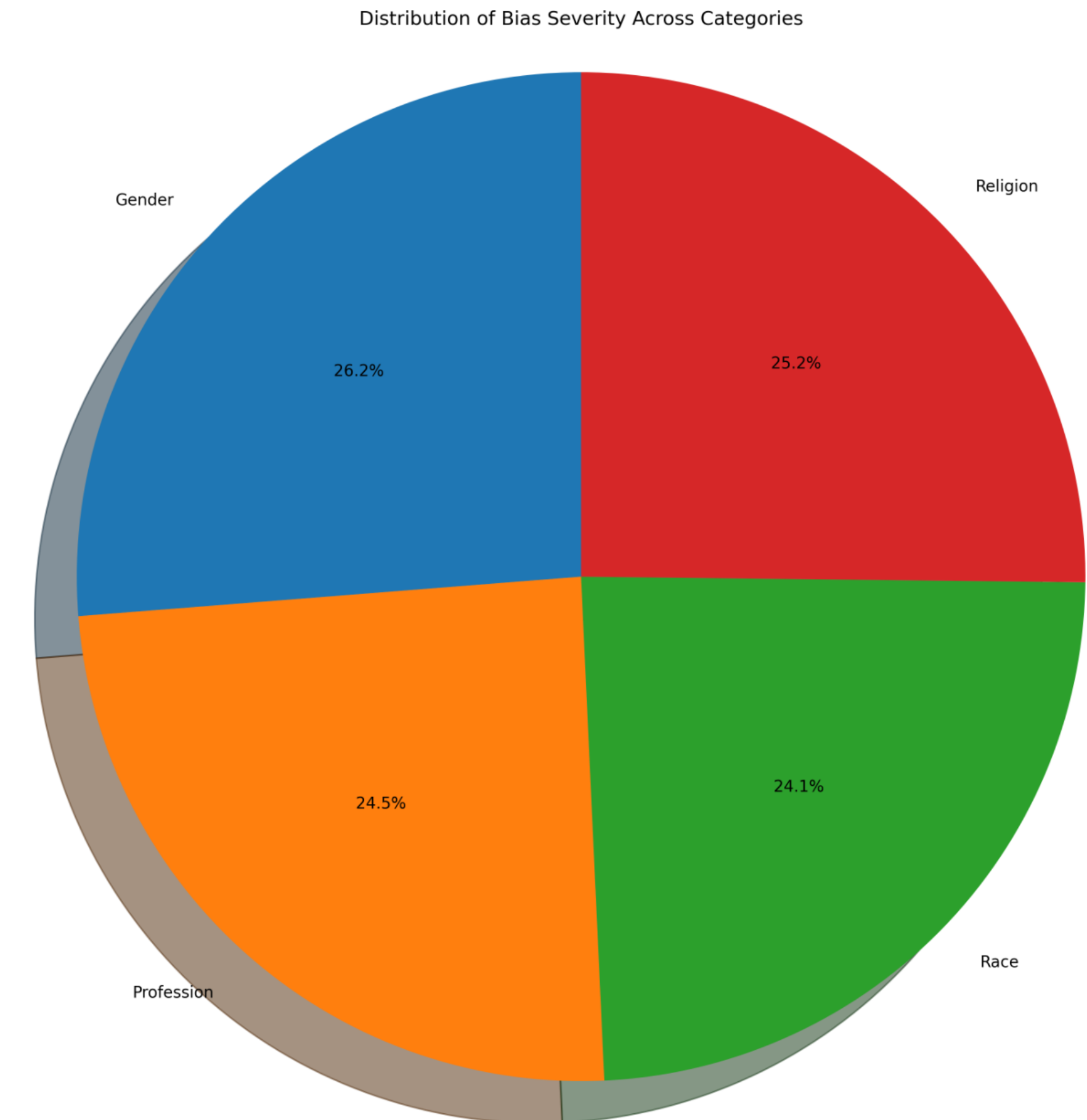
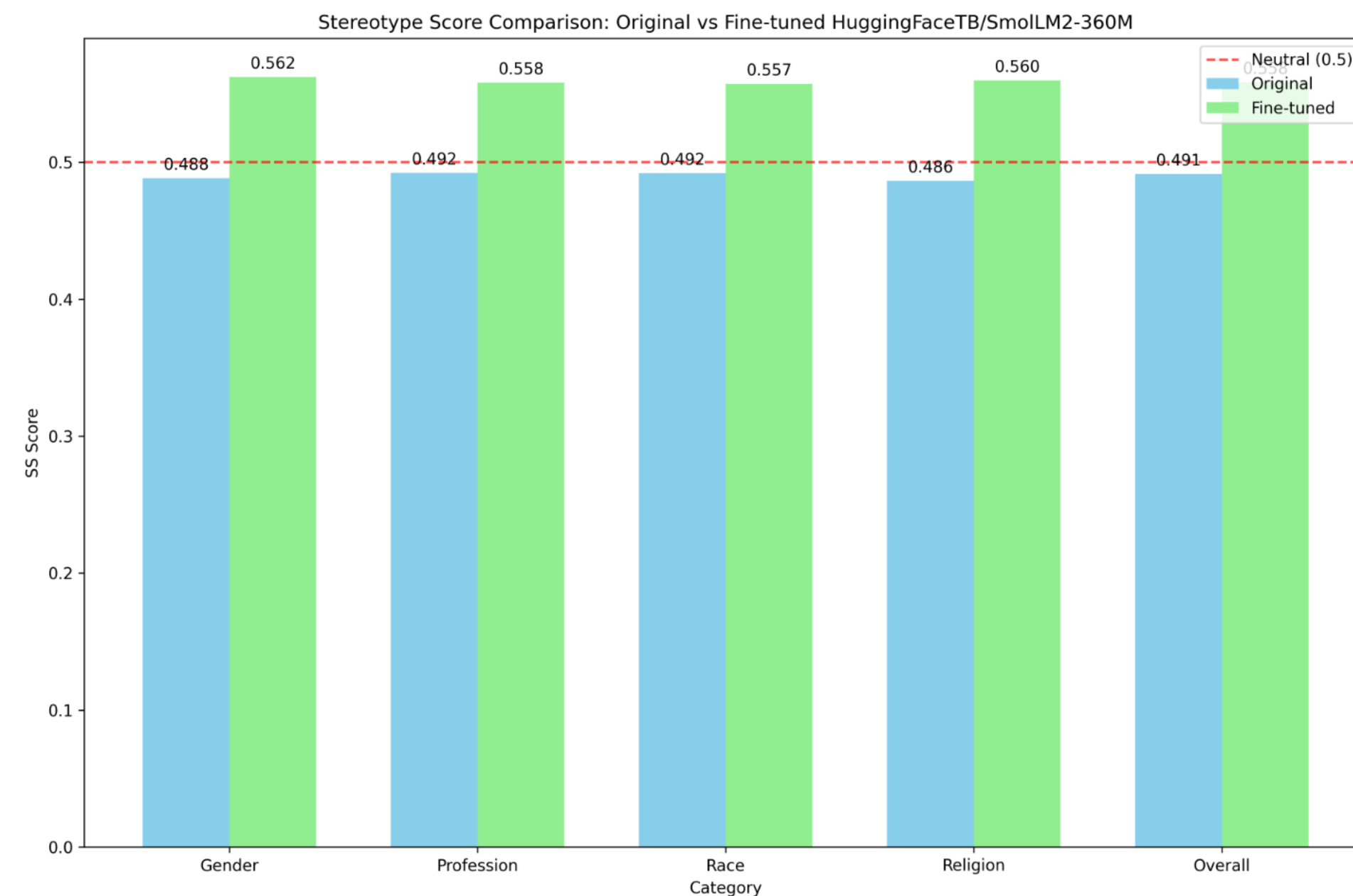
- Parameter Count: Maintain 360M parameter efficiency.
- Model Architecture: Preserve original causal language model structure.
- Fine-tuning Focus: Adjust token distributions while maintaining language understanding.

Post Fine-Tuning Results

Bias Direction Shift

- SS Score: 0.4915 $\rightarrow$ 0.5584 (+0.0669 change).
- Bias Direction: Anti-stereotype $\rightarrow$ Stereotype bias (over-correction).
- Bias Severity: Increased from 0.0085 to 0.0584.

Category-wise Changes



Analysis & Interpretation

Bias Direction Transformation

- Baseline State: Model exhibited anti-stereotype bias ($SS = 0.4915$).
- Post-Training: Shifted to stereotype bias ($SS = 0.5584$).
- Magnitude: $+0.0669$ change overshoots the ideal neutral point of 0.5.
- Consistency: All four bias categories showed similar over-correction pattern.

Implications for Bias Mitigation

- Effectiveness: Counter-stereotypical training successfully modifies bias direction.
- Precision Challenge: Achieving exact neutrality ($SS = 0.5$) requires careful calibration.
- Sensitivity: Models highly responsive to training data composition.

Significant Quality Improvements

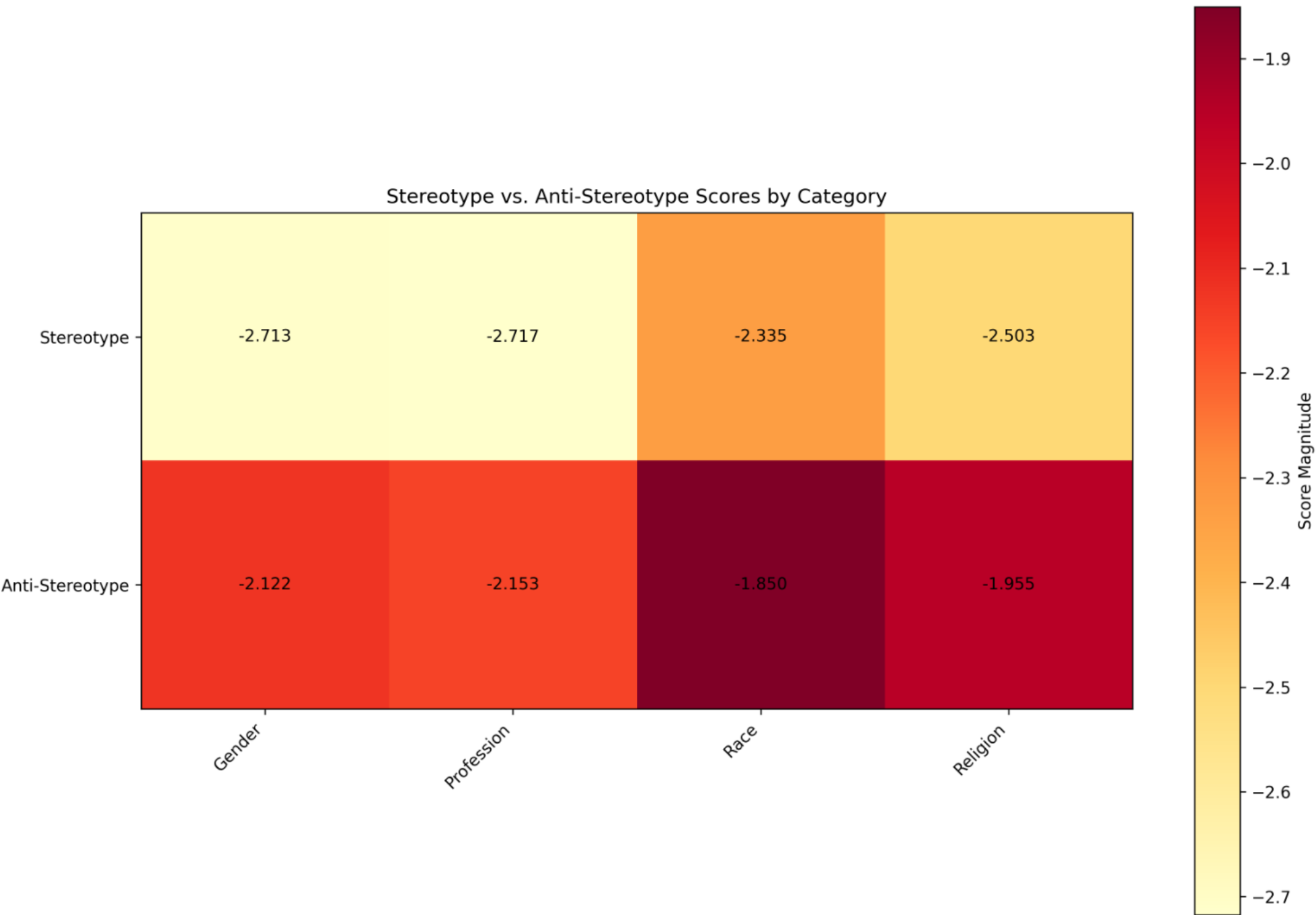
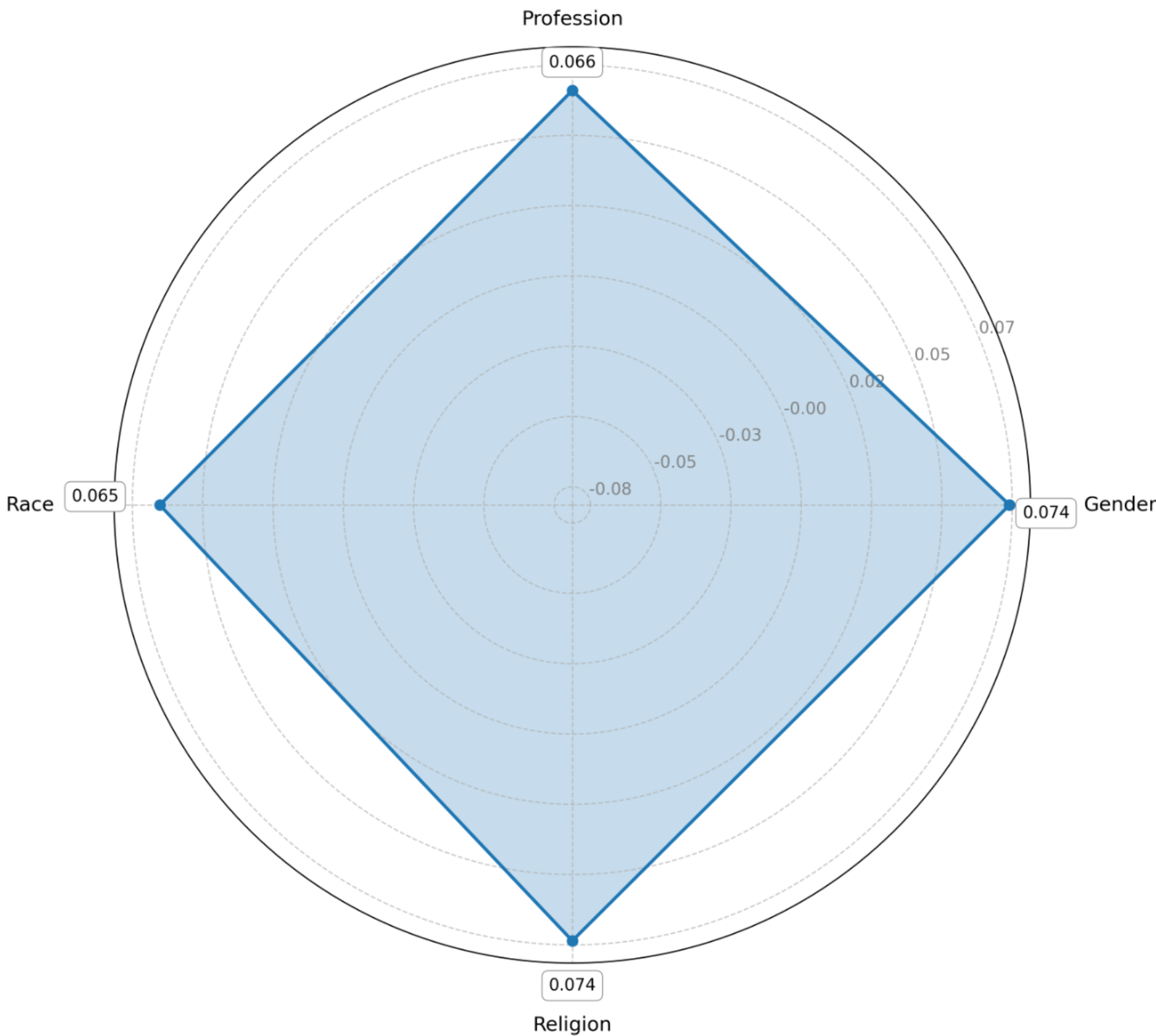
- Stereotype Scores: Improved from ~ -4.5 to ~ -2.5 range across categories.
- Anti-Stereotype Scores: Enhanced from ~ -4.7 to ~ -2.0 range.
- Overall Fluency: Substantial enhancement in language modeling capabilities.
- Performance Trade-off: Bias modification improved rather than degraded model quality.

Category-Specific Analysis

- Gender & Religion: Strongest over-correction ($+0.0740$, $+0.0736$).
- Profession & Race: Moderate but consistent over-correction ($+0.0659$, $+0.0653$).
- Pattern Consistency: No category-specific trade-offs or negative transfers.

Analysis & Interpretation

Improvement by Category for HuggingFaceTB/SmolLM2-360M



Conclusions

Primary Findings

- Bias Modification Success: Demonstrated ability to significantly alter bias patterns.
- Language Quality Preservation: Enhanced rather than degraded language modeling.
- Consistent Effects: Uniform improvements across all bias categories.
- Lightweight Efficiency: Maintained 360M parameter computational efficiency.

Critical Insights

- Over-Correction Challenge: Fine-tuning shifted model beyond ideal neutrality.
- Calibration Importance: Need for precise control in bias mitigation.
- Direction Sensitivity: Models highly responsive to counter-stereotypical training.



Thank You